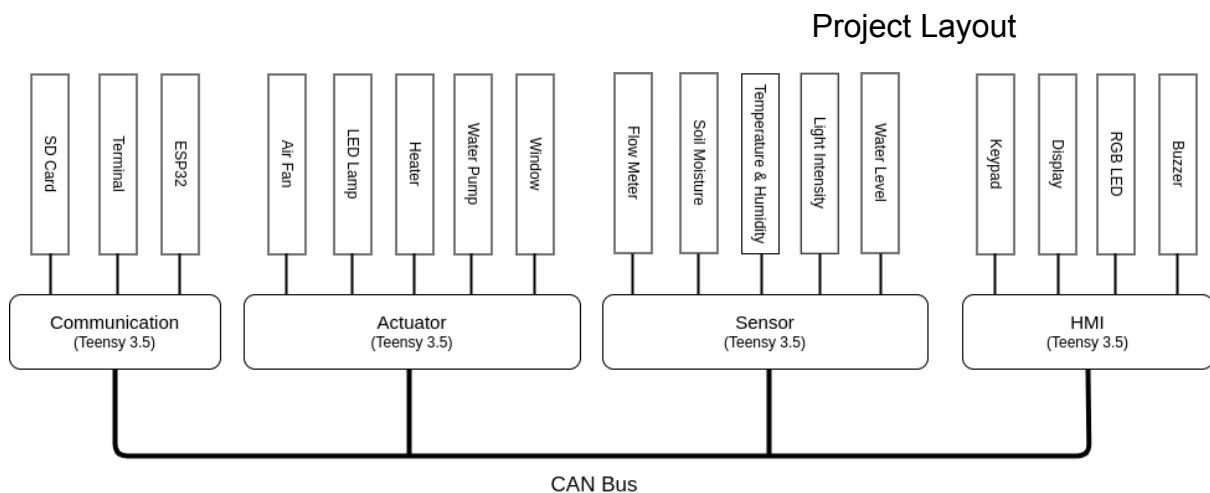
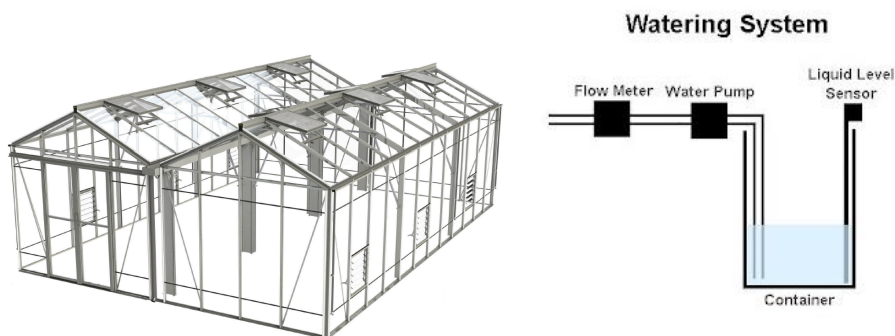


Teamwork Assignment

Requirements and Design

In this teamwork assignment you are supposed to develop requirements and design the software for an automated and connected greenhouse. The submission shall be a PDF file and in the submission there shall be:

- The stakeholder requirements of the greenhouse
- The system requirements, the architecture and the software design of the assigned node



Required Components

Name	No.	Link
Teensy 3.5	4	https://www.pjrc.com/teensy/K64P144M120SF5RM.pdf
ESP32	1	https://www.espressif.com/sites/default/files/documentation/esp32_technical_reference_manual_en.pdf
Display	1	https://www.elfa.se/Web/Downloads/_m/an/4311_enq_man.pdf
Keypad 3×4	1	https://www.sparkfun.com/products/14662
Buzzer	1	https://www.electrokit.com/uploads/productfile/41015/Passive_Piezo_Buzzer.pdf
Window	1	
Water Pump	1	https://www.kjell.com/se/produkter/el-verktyg/arduino/tillbehor/vatskepump-240-lh-p87079
Heating Element	1	https://www.conrad.se/p/thermo-tech-polyester-varmefolie-sjalvhaftande-12-vdc-12-vac-15-w-kapslingsklas-s-1px4-l-x-b-150-mm-x-60-mm-1216594
LED Lamp	1	https://www.clasohlson.com/se/LED-ljuslist-12-V-Northlight/p/36-5066
Air Fan	1	https://www.kjell.com/se/produkter/el-verktyg/elektronik/elektromekanik/flaktar/12-v/axialflakt-12-v-80x80x25-mm-p36175

Fluid Level Sensor	1	https://www.electrokit.com/uploads/productfile/41009/eTape_Datasheet_12110215TC-8_040213.pdf
Temp. & Humidity Sensor	1	https://www.electrokit.com/en/product/digital-temperature-and-humidity-sensor-dht11/
Light Intensity Sensor	1	https://www.electrokit.com/en/product/photo-resistor-cds-2-5-kohm/
Soil Moisture Sensor	1	https://www.electrokit.com/en/product/jordfuktighetssensor-kapacitiv-analog/
Flow Meter	1	https://www.electrokit.com/en/product/flodesmatare-1-30l-m-1-2-2/
Water Container	1	
Servomotor	1	https://www.mouser.se/ProductDetail/Parallax/900-00005?qs=sGAEpiMZZMsMLZWdhVL715W2VioMNnXk

Some Requirements

1. The temperature inside the greenhouse shall be kept in a configurable range using a temperature sensor, a heating element, an air fan and a window.
2. The humidity inside the greenhouse shall be kept in a configurable range using a humidity sensor, an air fan and a window.
3. The ambient light intensity inside the greenhouse shall be kept in a configurable range using a light intensity sensor and a LED lamp.
4. The soil moisture of the greenhouse shall be kept in a configurable range using a soil moisture sensor, an air fan, a window and the watering subsystem.
5. The container in the watering subsystem is manually filled with mixed water & fertilizer
6. The consumed water per watering shall be limited to a configurable volume.
7. In the HMI node It shall be possible to monitor, control and calibrate the system
8. The system shall operate safely and in the case of fatal failures it shall switch to a safe mode.
9. The HMI shall alert the user in the case of a fatal failure using a buzzer.
10. The HMI shall show the status of the system using different colors on an RGB LED.
11. The HMI shall store and restore the configuration values in/from the EEPROM of Teensy.
12. The HMI shall display the current date & time and the system status.
13. The system shall be connected to the Internet using an ESP32.
14. It shall be possible to monitor the system via MQTT and a broker
15. The system shall automatically synchronize date&time every night at 3 o'clock
16. It shall be possible to communicate to the system using a terminal in order to
 - a. provide date&time for the system
 - b. get snapshot of the system

Grading

Your work is evaluated and graded as a team.

The criteria for G(Godkánt) are:

- You collaborate actively in the development of the requirements, architecture and software design.
- On Tuesdays and Thursdays it is mandatory to be in the school.
- The requirements shall be good requirements.
- The architecture and the software shall be well-designed.